Duarte Cardoso

Porto, Portugal (open to relocation within Europe) | +351 927570952 | caldasdcardoso@gmail.com
linkedin.com/in/duartecardoso | github.com/duartecaldascardoso | github.com/complydoc/complydoc | portfolio & blog

*AI Engineer with 2+ years building production LLM and agentic systems for enterprise document workflows in insurance and finance, after 1.5 years as a full-stack engineer. Author and sole maintainer of **complydoc**, open-source observability for document ingestion pipelines.*

Experience

DareData Engineering — Porto, Portugal (hybrid)
AI Engineer — *Jul 2026 – Present*

- AI consultant for **Euronext**, designing, developing and deploying **cloud-native AI solutions on AWS and Azure** that automate operational workflows.
- Main developer of **FreeFloat**, a **Dataiku**-integrated **browser agent** (Strands, AWS) automating free float review across **614 companies**, targeting ∼120 person-days saved annually across four review cycles.

msg insur:it Iberia — Porto, Portugal (hybrid)
AI Engineer — *Aug 2024 – Jul 2026*

- Built the company's **official Document Intelligence tool** on an **Agentic RAG architecture**, serving ∼90 users company-wide since launch.
- Designed and developed **Configure:it**, a **multi-agent** Product Machine module converting documents into product configurations and enabling natural language querying.
- Owned the whole **validation engine** of Product Machine, managing an intern and delivering a **production-ready** implementation.
- Built one of its components as my in-house Master's thesis: the **Product Validation Accelerator Engine**, which combines **LLM agents** with **combinatorial testing**, boundary value analysis and equivalence partitioning to turn a plain-language request into a complete test suite, cutting insurance product validation from ∼6 months of manual client work to a handful of automated requests.
- Worked alongside two Master's thesis students as their buddy at work, on LLM-based extraction of insurance product characteristics and natural-language exploration of product models, leading to two co-authored papers at SEI 2025, one awarded **Best Paper**.

Software Engineer (Intern, Feb – Jun 2023) — *Feb 2023 – Aug 2024*

- Full-stack engineer on an enterprise-grade insurance web application for a North American client, in an international, cross-functional team (**Java Spring**, **Backbone.js**, **PostgreSQL**); contributed to a **go-live with a single bug** reported across the whole production setup.
- As an intern, built an end-to-end insurance product proof of concept, implementing workflows with **Camunda (BPMN/DMN)** integrated with Java services.

Projects

complydoc | *Python, CLI, GitHub Actions, LangChain, LlamaIndex, Docling* — GitHub | PyPI

- Sole author and maintainer of an **MIT-licensed observability layer for document ingestion pipelines**, published on PyPI: traces every loading, splitting and embedding step with **timing, token counts and cost**, so teams know what their documents contain before they reach an LLM.
- Puts each loader's reading side by side with the source page, **a diff for extracted documents**, so engineers can see lost headings, broken tables or dropped lines and choose loaders on their own documents (LangChain, LlamaIndex, Unstructured, Docling, LlamaParse, Azure Document Intelligence).
- Built `complydoc check`: **YAML policies** and a **GitHub Action** that gate ingestion in CI by flagging checksum-validated personal and financial identifiers, hidden text and **prompt-injection** passages, masked until reviewed. Runs fully locally, with outbound network blocked by default.

me-agent | *Python, LangChain, FastAPI, uv* — GitHub

- End-to-end **agentic RAG** system for personal documents: ingestion of PDF, DOCX and Markdown into a vector store, retrieval, and an agent answering with that context, served through a **streaming FastAPI** endpoint and a CLI.

snappy-diff | *Rust, JavaScript* — GitHub

- Fast side-by-side diff viewer for git branches, uncommitted changes and patch files, with keyboard navigation, file search and install scripts for macOS, Linux and Windows.

Article Explainer with Swarm Architecture | *Python, LangChain, Multi-Agent Systems* — GitHub

- Multi-agent explainer built on LangChain's Swarm pattern; **featured twice by LangChain on LinkedIn** as an exemplary implementation.

Education

Faculty of Engineering, University of Porto (FEUP) — Porto, Portugal
M.Sc. Software Engineering (18/20) — *Sep 2024 – Jul 2026*
- Thesis defended with **19/20**.

Polytechnic of Porto, School of Engineering (ISEP) — Porto, Portugal
B.Sc. Informatics and Computing Engineering (14/20) — *Sep 2020 – Sep 2023*

Publications

Simplifying Complex Insurance Product Management with AI — SEI 2025
***Best Paper**, co-author* — *Nov 2025*
- Product Machine Explorer, an LLM- and agent-powered tool to explore complex insurance product models through natural language. Proceedings.

Automated Extraction of Insurance Product Characteristics — SEI 2025
Co-author — *Nov 2025*
- An AI agent workflow, supported by NLP and LLMs, extracting insurance product characteristics from regulatory documents with structured, quality-assured outputs.

Writing

Write about LLMs, agents and document intelligence on my blog and LinkedIn (1,400+ followers), covering OCR versus vision LLMs for extraction, open-weight models in regulated industries and small decision models for extraction pipelines. A post on open-weight models reached **70k+ impressions**, and one on complydoc **14k+**.

Certifications

Claude with Amazon Bedrock (Anthropic, Mar 2026) · **MCP: Advanced Topics** (Anthropic, Feb 2026) · **ML Model Development** (Databricks, Feb 2026)

Leadership

NEI, ISEP Computer Science Student Association — Porto, Portugal
Vice President (Communication member, Oct 2020 – Aug 2021) — *Aug 2021 – Aug 2022*
- Co-managed a team of close to 40 members running technology and student-company events.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, Rust
LLM & Agents: Agentic RAG, multi-agent systems, MCP, prompt engineering, LangChain, LangGraph, Strands, FAISS, Qdrant
Evaluation & Observability: LangFuse, LangSmith, MLflow, token and cost tracing
ML: PyTorch, Scikit-learn
Cloud & Data: AWS, Azure, Dataiku, PostgreSQL
Engineering: FastAPI, Pydantic, Java Spring, Camunda, React, Docker, Terraform, GitHub Actions, CI/CD
Spoken languages: Portuguese (native), English (fluent), Spanish (proficient)

Interests

Running and triathlon: ran the Porto Half Marathon (Sep 2026); training for the Porto Marathon and Ironman 70.3 Vilamoura.